

SHAIK MOHAMMAD ABRAR AHAMMAD

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Education

BS Abdur Rahman Crescent Institute of Science and Technology

Chennai, India

Bachelor of Technology – Artificial Intelligence and Data Science; CGPA: 8.87/10

Jul 2022 – Apr 2026

Sri Chaitanya Junior College

Vijayawada, India

Intermediate (12th Grade); Percentage: 94.8%

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, R, SQL

AI/ML: LangGraph, LangChain, Scikit-learn, Pandas, NumPy, XGBoost, LightGBM, PaddleOCR, Sentence Transformers, RAG, Multi-Agent Systems, Prompt Engineering, NLP

Frameworks: React, Node.js, Express.js, Flask, FastAPI, .NET

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Git, CI/CD

Databases & Tools: ChromaDB, SQLite, PostgreSQL, MongoDB, REST APIs, Server-Sent Events (SSE)

Concepts: Data Analysis, Generative AI, Large Language Models, Vector Databases, Distributed Systems, Agile

Projects

NVIDIA-MCP-Swarm: Multi-Agent AI Collaboration Framework

2025

Python, FastAPI, React, LangGraph, ChromaDB

- Architected a full-stack multi-agent AI system utilizing the **Model Context Protocol (MCP)** with a **Rotational LLM Council** that randomly selects 4 of 7 heterogeneous NVIDIA NIM models per query for consensus-driven responses.
- Designed a **LangGraph StateGraph** orchestration pipeline with typed state management, conditional routing, and parallel execution, achieving a **22% improvement in task completion** and **11.5% reduction in hallucination rates**.
- Implemented **RAG-based persistent vector memory** using ChromaDB and Sentence Transformers (384-dim embeddings), reducing context window redundancy by **40%** and enabling **91% cross-session fact recall**.
- Built three MCP primitives – **Resources**, **Prompts**, and **Tools** – enabling IDE integration (VS Code, Cursor) and standardized agent interoperability via JSON-RPC 2.0.
- Developed a production-ready React frontend with real-time SSE streaming, council dialogue visualization, document intelligence (Nemotron-Parse), persistent conversation history, and a memory toggle for adaptive performance.

MIRSAD: AI-Powered Energy Supply Chain Resilience Platform

2026

Python, FastAPI, LangGraph, Gemini 2.0, RAG, Leaflet.js

- Developed a real-time geopolitical risk monitoring and disruption scenario modeling platform for India's crude oil security, integrating live news feeds and supply data into a centralized Command Center.
- Engineered a 9-node **LangGraph** AI pipeline with **RAG-grounded** analysis utilizing **Gemini 2.0 Flash** to generate quantitative risk scores, predictive 30-day risk trajectories, and deterministic supply security metrics (SSI, HHI).
- Built interactive disruption war-gaming simulations to model cascading economic impacts, dynamically re-ranking alternative procurement sources and optimizing Strategic Petroleum Reserve (SPR) drawdown strategies.

Real-Time Competitive Programming Battle Platform

2025

React, Express.js, AWS, WebSocket

- Designed and developed a scalable, real-time coding battle platform (LeetCode-style), enabling matched users to compete on competitive programming problems with dynamic, chess-style Elo rating updates.
- Engineered real-time matchmaking using WebSockets, secure user authentication, and **AWS-hosted** scalable infrastructure for concurrent multi-user sessions.